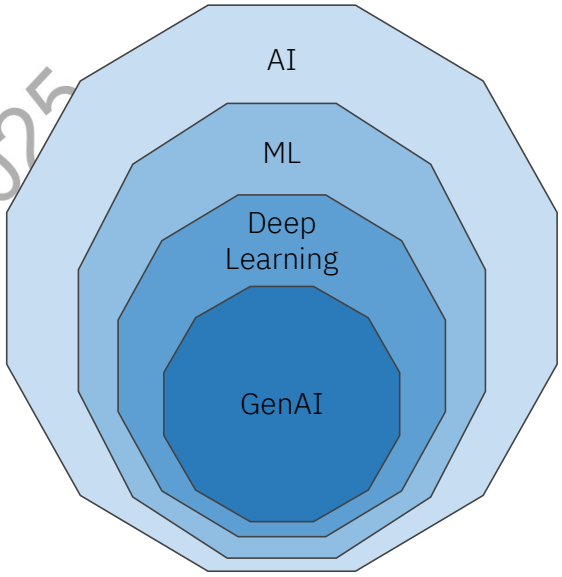
Introduction to AI

Week 6: Unsupervised machine learning

Nguyen Tuan Khang *PhD.* | khangnt@vn.ibm.com |
090 830 6668
Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

USTH 2025



1

Objectives

- Differentiate supervised and unsupervised learning in terms of data requirements, objectives, and real-world use cases.
- Explain the core concepts behind key unsupervised methods, including Clustering (K-Means) & Dimensionality Reduction (PCA)
- Labs: Iris Dataset

USTH 2025

2

2

AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation and logic
W3: Searching algorithms (pathfinding and text)

Machine Learning

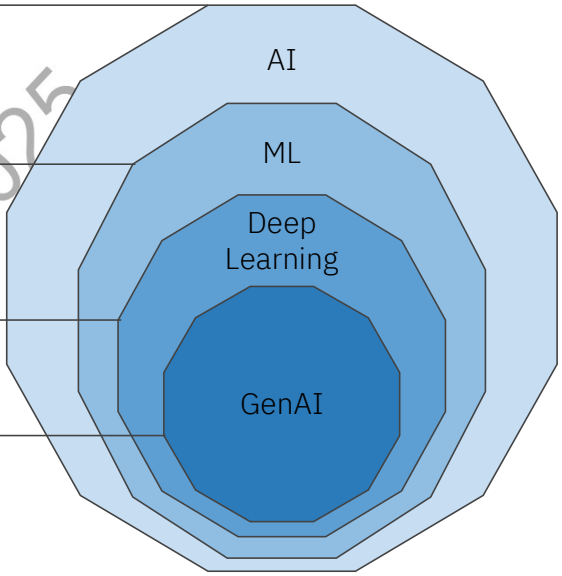
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



USTH 2025

3

3

Contents

- Unsupervised Machine Learning
  - Feature Engineering – Basic Techniques
  - Clustering: K-Means
  - Dimensionality Reduction: PCA

USTH 2025

4

4

Data Labelling

- The Supervised Learning is powerful but **requires humans to label data.** That is expensive and slow.
- But in the real world, 90% of data has no labels. We need Unsupervised Learning to unlock that 90%

USTH 2025

10% Labeled Data

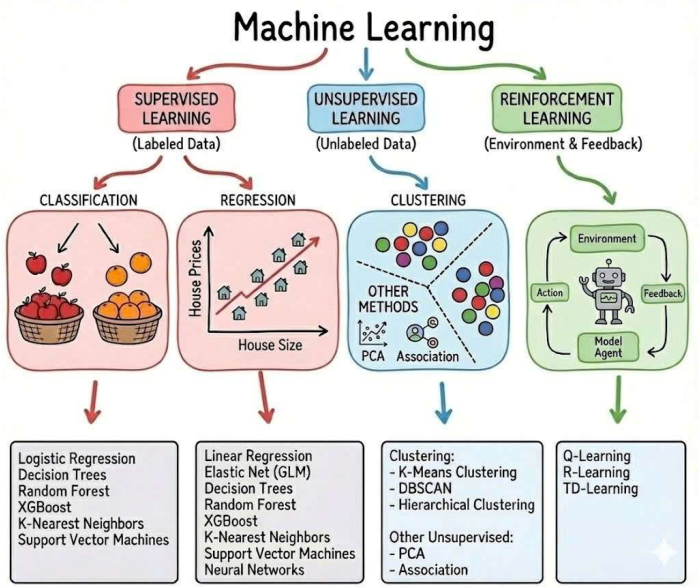


90% Unlabeled Data

5

5

Categories of ML



6

6

Unsupervised Learning

Formal Definition

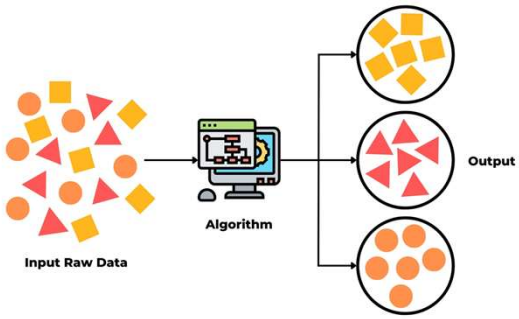
Unsupervised learning is a ML in which a model is trained on unlabeled data x_i ,with the objective of discovering hidden structure, patterns, or relationships within the dataset.

Approach

- No labels (y) are provided; the algorithm infer structure autonomously.
- The results are evaluated based on statistical properties, such clustering, feature learning or dimensionality reduction
- The goal may include clustering, dimensionality reduction, density estimation, or feature learning.

Examples

- Clustering: customer segmentation (K-means)
- Dimensionality reduction: PCA for visualization
- Anomaly detection: identifying outliers



Reference

Pattern Recognition and Machine Learning (Bishop)

USTH 2025

7

7

Supervised vs Unsupervised (Definition)

Aspect	Supervised Learning	Unsupervised Learning
Data	Labeled (x, y)	Unlabeled (x)
Goal	Learn mapping f(x)→y	Discover structure in data
Evaluation	Prediction accuracy	Statistical quality (e.g., cluster cohesion)
Typical Methods	Logistic Regression, Trees, SVM	K-means, PCA, Autoencoders

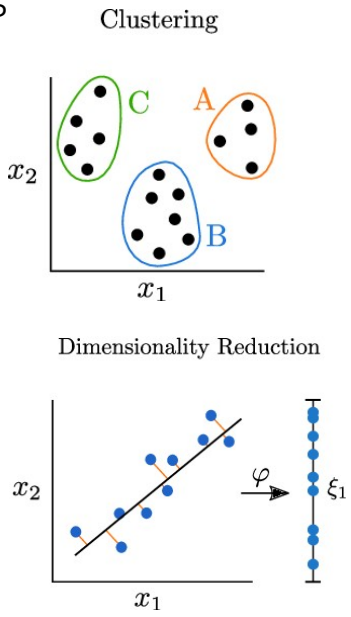
USTH 2025

8

8

Two primary goals of Unsupervised Learning

- Clustering
  - Group similar data points together
  - Each group represents a hidden pattern or behavior
- Dimensionality Reduction
  - Reduce the number of features
  - Keep the most important information
  - Make data easier to understand and process



USTH 2025

9

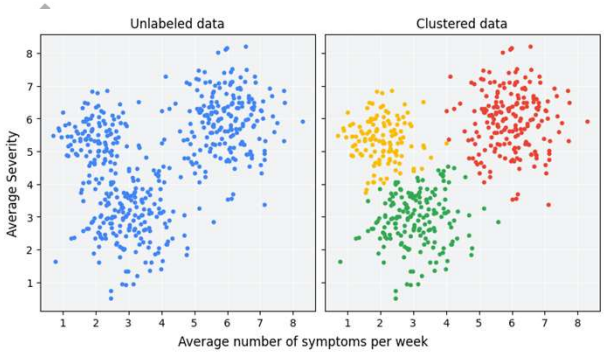
9

Clustering

Clustering is an unsupervised learning technique that groups observations into clusters based on similarity, so that items in the same cluster are more similar to each other than to those in other clusters.

Real-World Examples:

- Customer segmentation in marketing based on buying behavior
- Document grouping (news articles, emails, search results)
- Image segmentation by grouping similar pixels



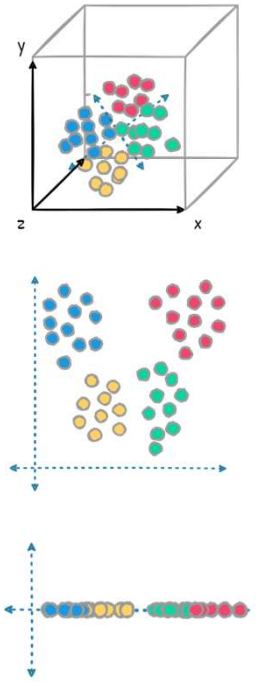
Source: https://developers.google.com/machine-learning/clustering/overview

10

10

Dimensionality Reduction

Dimensionality Reduction is transforming high-dimensional data into a lower-dimensional space so that the lower-dimensional representation retains meaningful properties of the original data.



Real-World Examples:

- Data visualization (reducing many features to 2D or 3D plots)
- Feature compression to speed up machine learning models
- Noise reduction in images, signals, or sensor data

USTH 2025

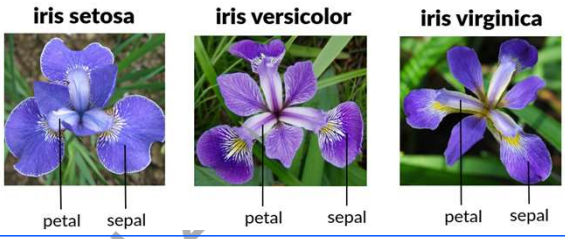
11

11

Case Study: Unsupervised learnings with Iris Dataset

Dataset

The Iris Dataset
(150 samples, 4 features, 3 labeled species)



Clustering

Problem
- Cluster the raw data into distinct groups without using species labels

Features (4)
- Sepal length/ width
- Petal length/ width

Model
- K-Means Clustering

Dimensionality Reduction

Problem
- Reduce the dimensionality (from 4 to lower) then visualize it on a 2D or 3D space

Features (4)
- Sepal length/ width
- Petal length/ width

Model
- Principal Component Analysis (PCA)

USTH 2025

12

12

Contents

- Unsupervised Machine Learning
- Feature Engineering – Basic Techniques
- Clustering: K-Means
- Dimensionality Reduction: PCA

USTH 2025

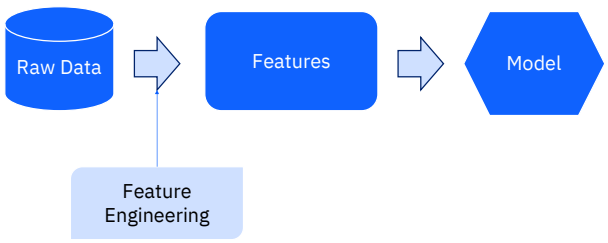
13

13

Feature Engineering Overview

- Raw data is not ready for ML
  - Machine learning models **cannot directly understand** text, categories.
  - Features may have **different units and scales**, which can bias the learning process.
  - Some features may be **highly correlated or redundant**, adding noise instead of value.

Feature Engineering transforms raw data into a form that models can learn from effectively.



USTH 2025

14

14

The Objectives of Feature Engineering

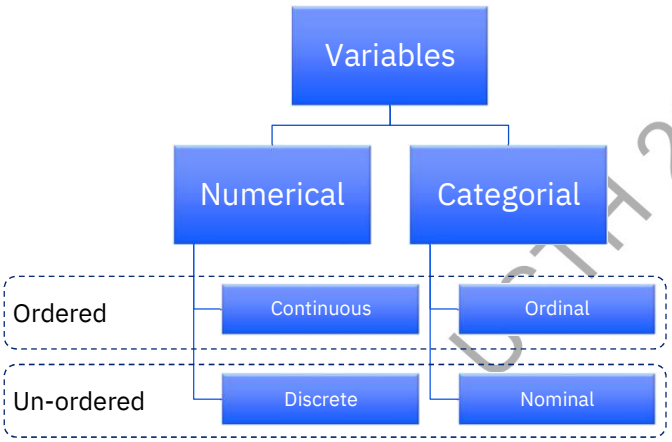
- Well-prepared features lead to better models
  - Faster and more stable model training
  - Improved accuracy and performance
  - Better interpretability and more reliable results
- Basic techniques (in this lesson, there are more)
  - Encoding (Label Encoding, One-Hot Encoding)
  - Feature Scaling (Normalization, Standardization)
  - Correlation Analysis
  - Dimensional Reduction

USTH 2025

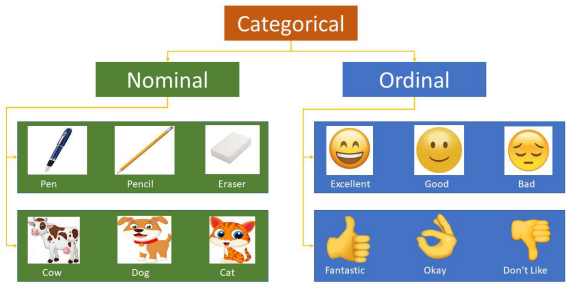
15

15

Variables (Data Types)



Examples of Categorial data (Source: LinkedIn)



USTH 2025

16

16

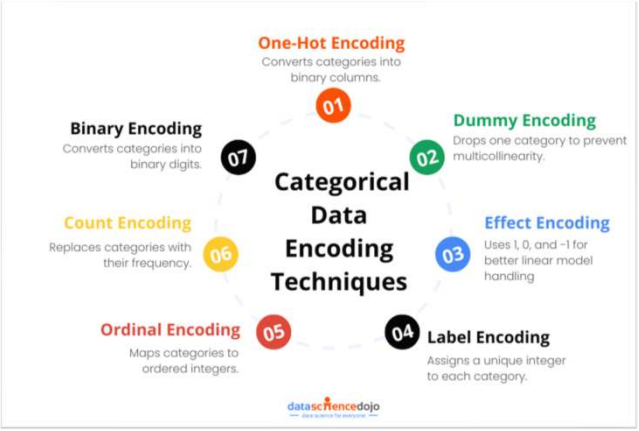
Encoding

Why Is Encoding Needed ?

- ML algorithms require numerical input
- Categorical data (e.g., “Spam/Not Spam”) need converted to numbers

Common Encoding Methods

- Label Encoding – assigns integers to categories
- One-Hot Encoding – creates binary features
- And more …



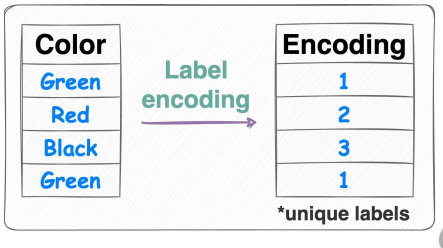
https://datasciencedojo.com/blog/categorical-data-encoding/

17

17

Encoding Techniques: Label vs One-hot Encoding

Label Encoding



One-hot Encoding

color	color_red	color_blue	color_green
red	1	0	0
green	0	0	1
blue	0	1	0
red	1	0	0

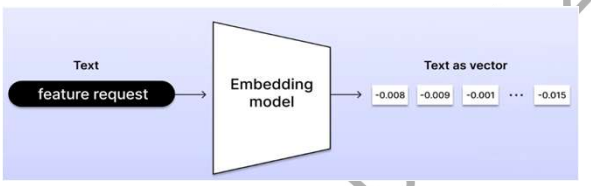
- **Increases dimensionality** when many categories
- Produces **sparse vectors** (most zero values)
- Does not capture similarity or **meaning between categories**

18

18

Encoding is evolved into Embedding

- Encoding has evolved into Embedding by learning dense numerical representations that capture meaning and similarity, instead of using fixed, manually designed encodings.



One-hot encoding

request →	1	0	0	0	0
feature →	0	1	0	0	0
how →	0	0	1	0	0
issue →	0	0	0	1	0
credit →	0	0	0	0	1

Learned embedding

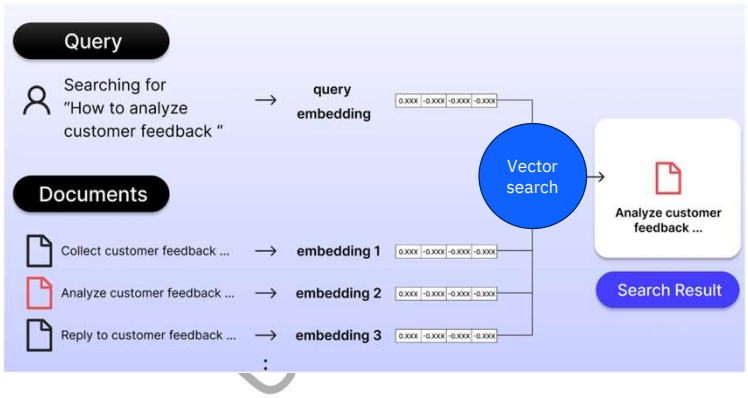
request →	0.007	-0.039	-0.002	-0.024
feature →	-0.014	-0.011	0.014	-0.017
how →	0.01	0.004	0.006	-0.037
issue →	-0.015	-0.035	-0.009	-0.01
credit →	-0.012	-0.027	-0.001	-0.05

https://syncly.app/blog/what-is-embedding-and-how-to-use

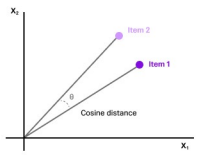
19

19

Semantic Search using Embedding



* Vector search can use cosine distance



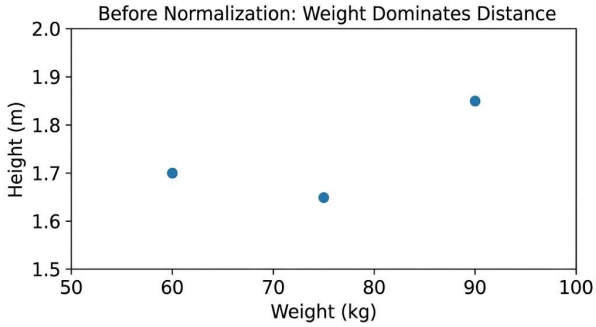
https://syncly.app/blog/what-is-embedding-and-how-to-use

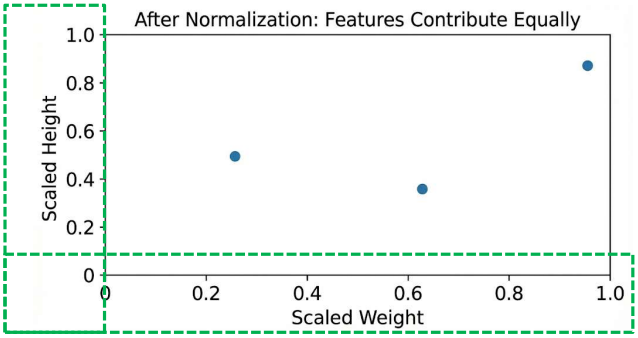
20

20

Feature Scaling

- Distance-based models compare features numerically
- Features with larger values can dominate distance calculations
- E.g., Measuring Weight in kilograms will dominate the distance Height in meters





USTH 2025

21

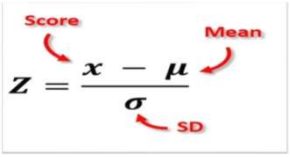
21

Common Feature Scaling Methods

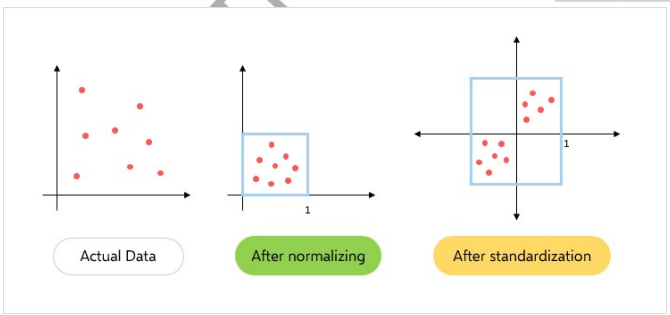
- Normalization (Min-Max scaling)
  - Focus on distance
  - The data is scaled [0,1]

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- Standardization (Z-score)
  - Focus on distribution
  - The data is scaled [-1,1]



$$z = \frac{x - \mu}{\sigma}$$

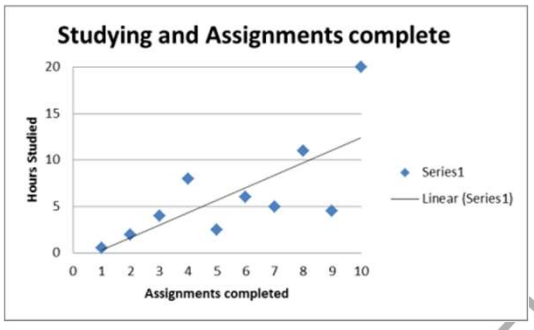


https://becominghuman.ai/what-does-feature-scaling-mean-when-to-normalize-data-and-when-to-standardize-data-c3de654405ed

22

22

Feature Correlation



USTH 2025

Why it matters

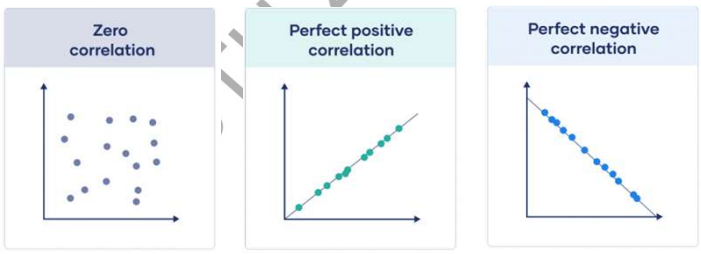
- Redundant features distort distance calculations, add up complexity.
- Strong motivation for Dimensionality Reduction

23

23

Correlation

- Correlation measures how strongly two numerical features move together
- Highly correlated features contain redundant information



24

24

Correlation Formulas

Correlation Coefficient

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Where:

x_i, y_i	• Individual data points of variables x and y
$\bar{x}, \bar{y}$	• How far each value is from its mean (centered data)
$(x_i - \bar{x}), (y_i - \bar{y})$	• Measures whether x and y move **together or oppositely**
$\sum(x_i - \bar{x})^2, \sum(y_i - \bar{y})^2$	• Measure the **spread (variance)** of each variable

25

25

Contents

- Unsupervised Machine Learning
- Feature Engineering – Basic Techniques
- Clustering: K-Means
- Dimensionality Reduction: PCA

26

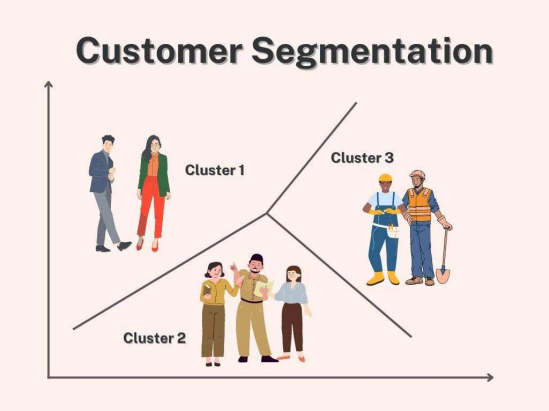
26

USTH 2025

Clustering Use Cases

Customer Segmentation:

- Clustering groups customers by behavior or spending patterns, …
- Help design targeted marketing, personalized offers, and better services —> without needing predefined labels.



More:

- Email grouping (similar emails without labels)
- Image compression (group similar colors)
- Document clustering (news articles, search results)

USTH 2025

27

27

Definition & Terminology

K-Means Clustering is an unsupervised learning algorithm that partitions data into K clusters by assigning each data point to the nearest cluster center (centroid), minimizing the distance within each cluster.

USTH 2025

- **Cluster:** A group of data points that are similar to each other
- **K:** The number of clusters to find

- **Centroid:** The center point of a cluster (average of all points in the cluster)
- **Distance Metric:** A way to measure similarity between points

28

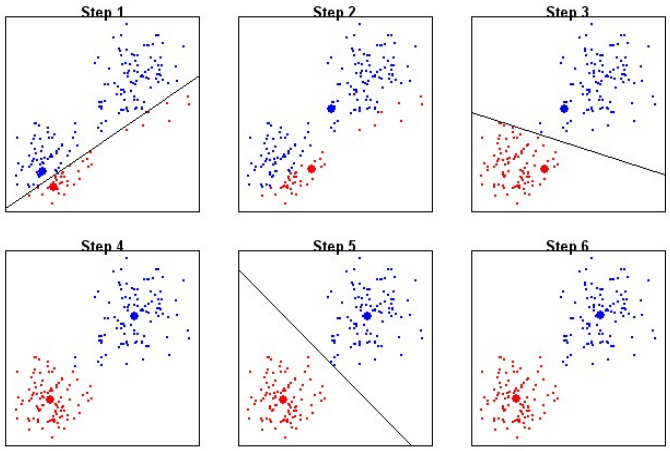
28

Clustering Process using K-means

Init: Choose K (number of clusters)

Process

1. Initialize centroids (randomly)
2. Assign points to the nearest centroid
3. Update centroids (mean of assigned points)
4. Repeat step 2 & 3 until clusters stop changing (convergence)



USTH 2025

29

29

Centroid Update Formula

After data points are assigned to clusters, the centroids are recalculated by taking the mean of all points in that cluster:

$$\mu_j = \frac{1}{|S_j|} \sum_{x \in S_j} x$$

USTH 2025

Where:

- $(|S_j|)$: *The number of points in cluster*
- $(j).(\sum x)$: *The sum of the coordinate vectors of all points assigned to that cluster.*

30

30

Loss Function

Within-Cluster Sum of Squares (WCSS) measures the total squared distance between each data point and the centroid of the cluster.

$$Loss = \sum_{i=1}^{n} \min_{j=1}^{k} |x_i - c_j|^2$$

Where:

- (n) is the total number of data points.
- (k) is the number of clusters.
- $(x_i)$is the $(i) - th$ data point.
- (c_j) is the centroid (mean) of the $(j) - th$ cluster
- $(|x_i - c_j|^2)$is the squared Euclidean distance between the data point $(x_i)$and the cluster centroid (c_j).

31

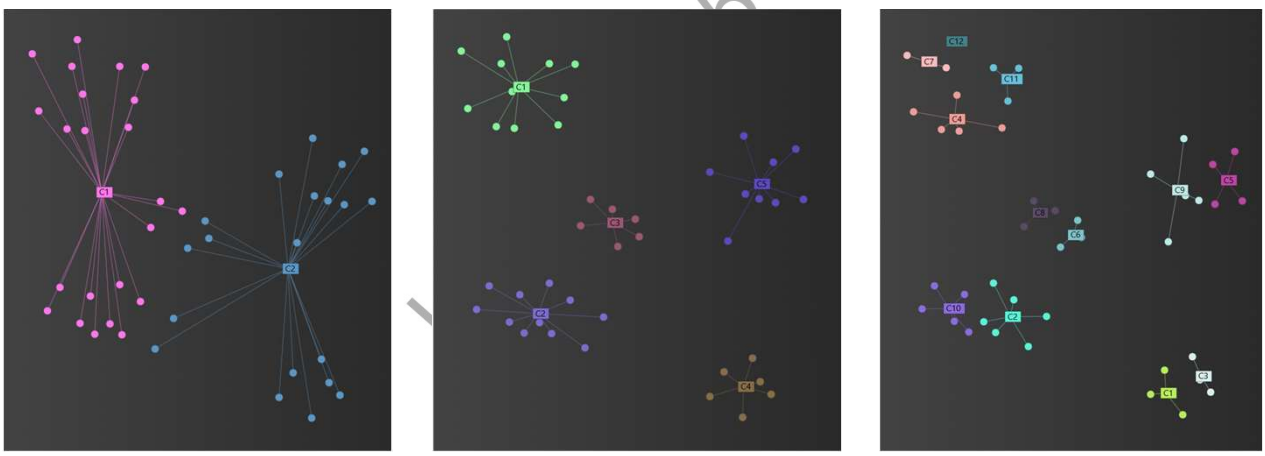
31

K-mean Simulation (Choosing K)

K too small → clusters are too broad (under-clustering)

Balanced K

K too large → clusters are fragmented (over-clustering)



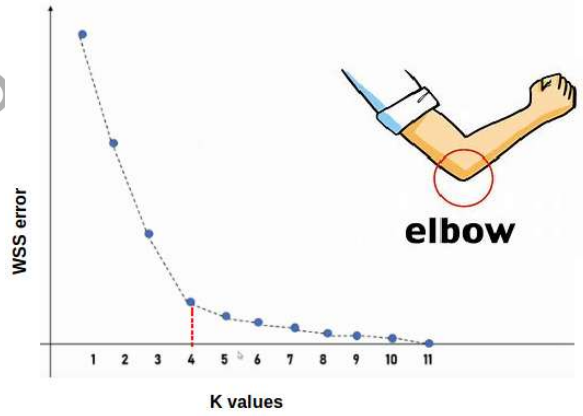
https://clustering-visualizer.web.app/kmeans

32

32

The Elbow Method to choose K

- A heuristic method to help choose a suitable value of K
  1. Run K-Means with different values of K
  2. Compute Loss for each K
  3. Plot Loss versus K as Elbow diagram
- The "elbow point" is where the rate of decrease slows down significantly
- This point represents a good trade-off between cluster quality and model complexity



USTH 2025

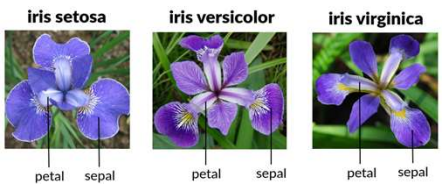
33

33

Lab 6.1: Clustering with Iris Dataset

Dataset

The Iris Dataset



Clustering

Problem

- Cluster the raw data into distinct groups without using species labels

Features (4)

- Sepal length/ width
- Petal length/ width

Model

- K-Means Clustering

Lab Explained (about 45')

- Step 1: Lab Setup
- Step 2: Perform Exploratory Data Analysis (EDA)
- Step 3: Feature Engineering
- Step 4: Model Training & Optimization
- Step 5: Outcomes

https://colab.research.google.com/drive/1BrERCOQxWArQZeLw677lz8Vaxns1DPod?usp=sharing

USTH 2025

34

34

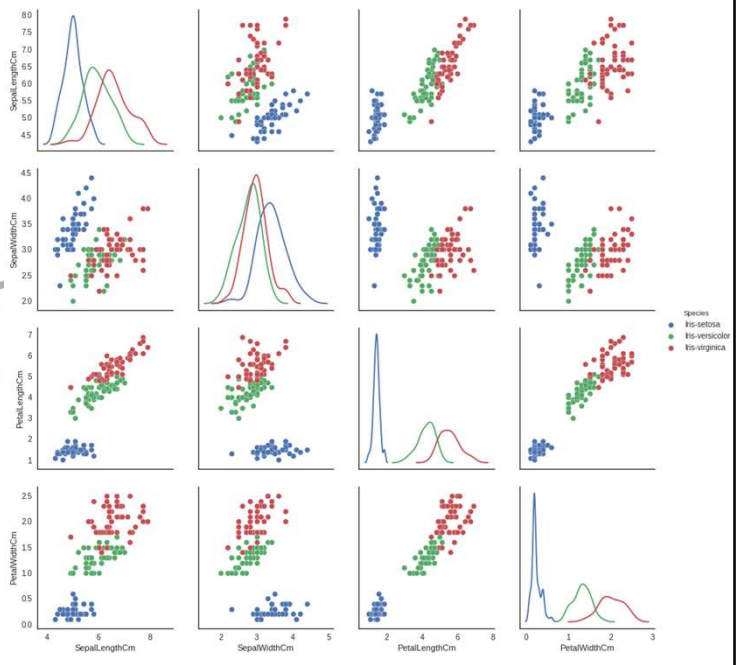
Dataset Introduction

The Iris dataset, introduced by statistician Ronald Fisher in 1936.

One of the earliest and most iconic datasets in statistics and machine learning, widely used as a foundational benchmark for classification, clustering, and dimensionality reduction.

```
Iris-virginica     50
Iris-versicolor    50
Iris-setosa        50
Name: Species, dtype: int64
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

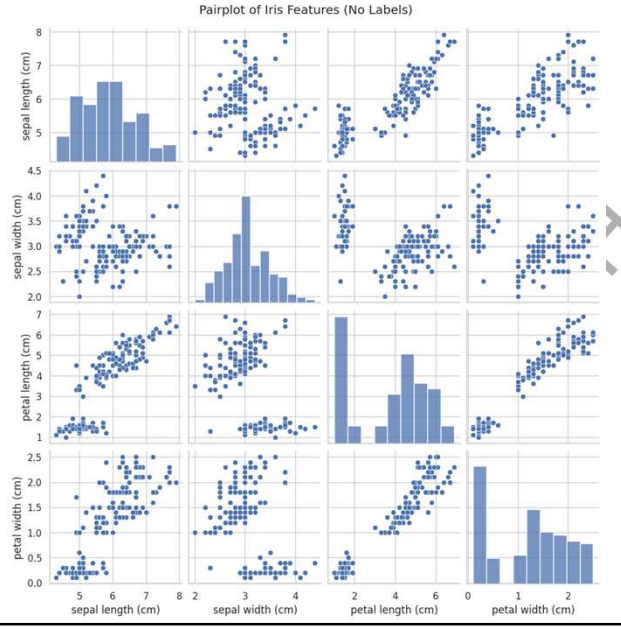


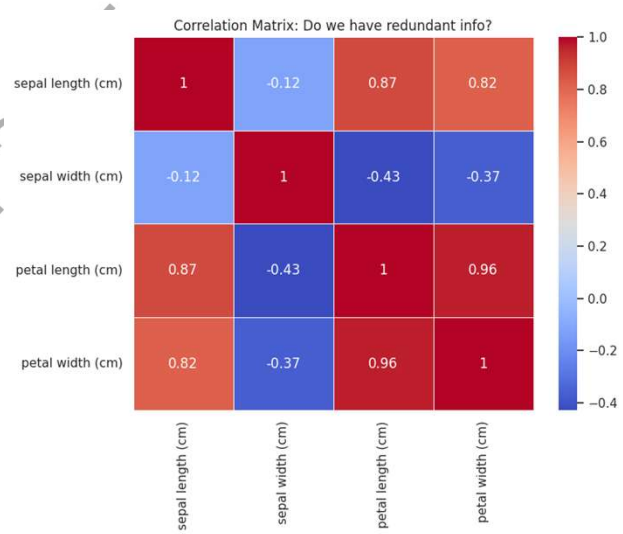
35

35

Lab 6.1 – Perform Exploratory Data Analysis (EDA)

Plot to see the relationship of every feature pair, calculate the correlation of all features





36

36

Lab 6.1 – Feature Engineering

Scale the data using Standardization scaler

```
# 1. Initialize Scaler
scaler = StandardScaler()

# 2. Fit and Transform
scaled_features = scaler.fit_transform(df)

# Check the new values (Mean should be ~0)
print("First 5 rows of Scaled Data:")
print(pd.DataFrame(scaled_features, columns=df.columns).head())
```

```
First 5 rows of Scaled Data:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0          -0.900681          1.019004          -1.340227         -1.315444
1          -1.143017         -0.131979          -1.340227         -1.315444
2          -1.385353          0.328414          -1.397064         -1.315444
3          -1.506521          0.098217          -1.283389         -1.315444
4          -1.021849          1.249201          -1.340227         -1.315444
```

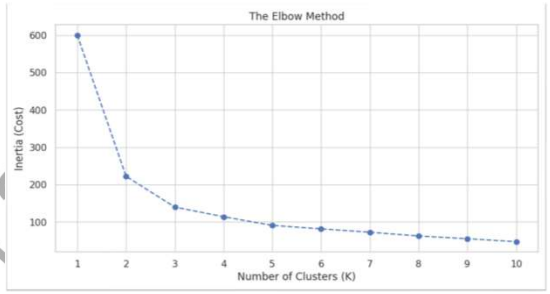
Since K-Means calculates similarity using Distance, scaling ensures that features with larger numeric ranges do not unfairly dominate the grouping process.

37

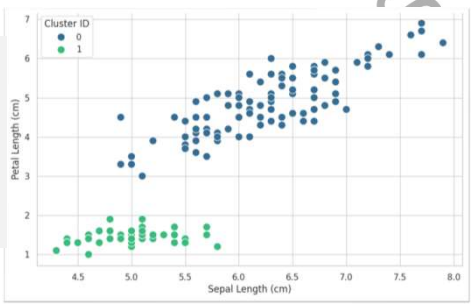
37

Lab 6.1 – Model Training & Optimization

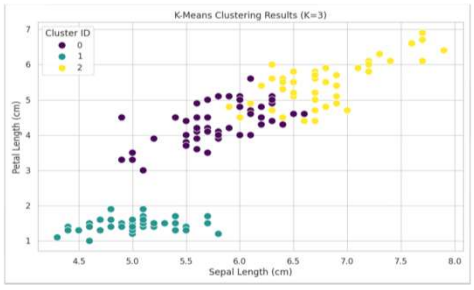
- Repeat the model with different K values (from 1 to N) and get corresponding Loss
- Draw the Loss diagram with K values to select the best K value



Example: k = 2



Example: k = 3



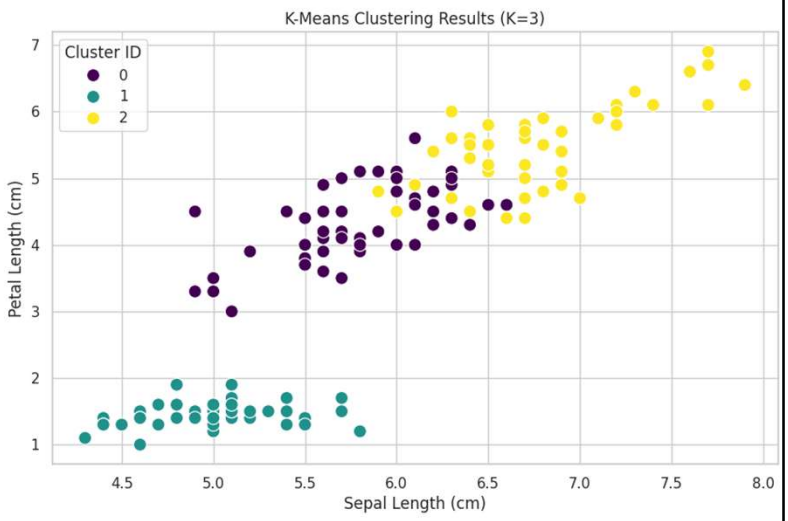
38

38

Lab 6.1 – Outcomes

Takeaways :

- The "Elbow Method" correctly predicted 3 groups, aligning perfectly with the real species (truth labels).
- Using only distance (no labels), the model discovered correctly the natural biological categories.

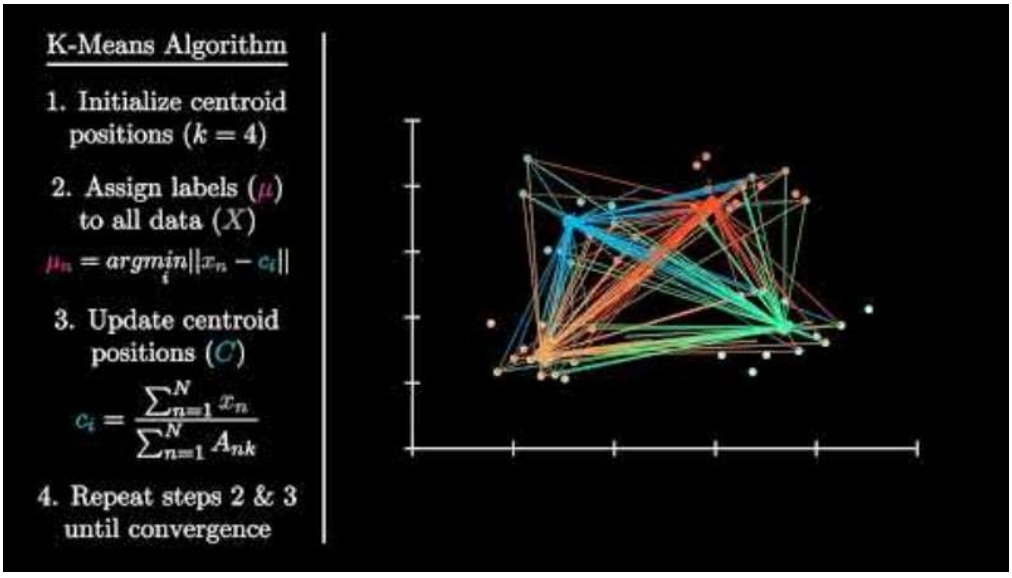


USTH

39

39

K-means Video



https://www.youtube.com/watch?v=2lZZ_FzlIJY

40

40

K-means Clustering — Key Takeaways

Pros

- Simple and intuitive to understand
- Fast and scalable for large datasets
- Effective when clusters are well-separated

Cons

- Must choose K in advance
- Sensitive to feature scaling and outliers
- Assumes clusters are roughly spherical

USTH 2025

41

41

Contents

- Unsupervised Machine Learning
- Feature Engineering – Basic Techniques
- Clustering: K-Means
- Dimensionality Reduction: PCA

USTH 2025

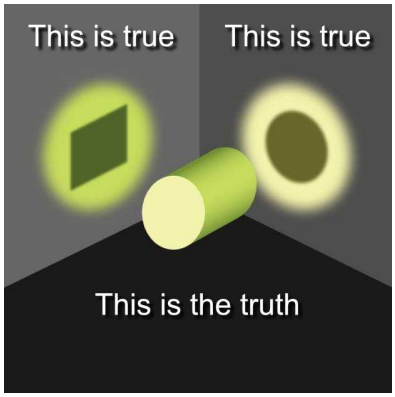
42

42

Dimensionality Reduction Explained

Data projection from different angle

- Imagine taking a photo (2D image) of a 3D object.
- From one angle, the shape looks clear.
- From another angle, important details are hidden.



What Dimensionality Reduction does

- Finds the best viewing angles of the data
- Projects data onto directions with maximum variation
- Discards angles that add little new information

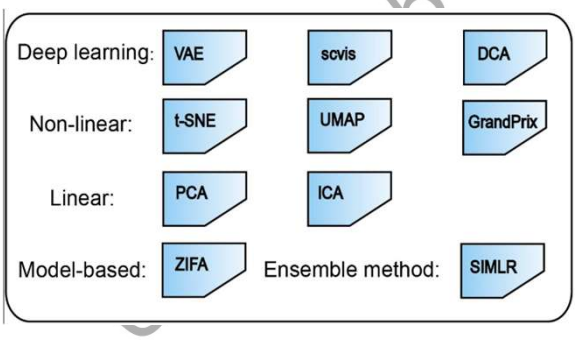
-> Dimensionality Reduction is to chooses the viewpoints that reveal the most information.

USTH 2025

43

43

Dimensionality Reduction Methods



https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2021.646936/full

44

44

PCA Definition & Terminology

Principal Component Analysis (PCA) is a technique that transforms high-dimensional data into a lower-dimensional space by creating new orthogonal features (principal components) that capture the maximum variance in the data.

- **Projection**: The process of mapping data points onto the principal components (PCx)
- **Principal Component**: New feature that represents direction of maximum variance in data

- **Eigenvector (Component Direction):** The direction along which the data varies the most
- **Eigenvalue (Variance)**: The amount of variance captured by a principal component
- **Explained Variance Ratio**: The proportion of total variance explained by each component
- **Reconstruction Error:** The information lost when reducing data to fewer dimensions

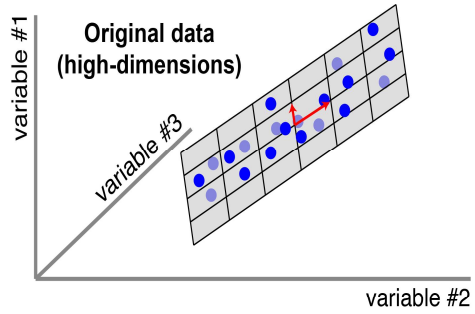
45

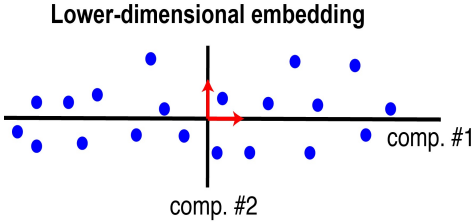
45

PCA Process

1. Standardize features (important for fair comparison)
2. Find directions of maximum variance in the data
3. Create principal components (new features)
4. Project data onto the top components
5. Keep only the most informative components

PCA creates new features — not select existing ones.





46

46

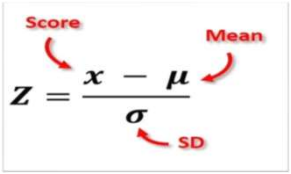
PCx Calculation

What is PCx ?

- PCx is the new feature obtained by projecting data onto the x-th principal component

How to Calculate PCx ?

- Scale the data with standardization (Xcentered)
- Compute *eigenvectors* x of the covariance matrix
- Project data onto the eigenvector to get PCx



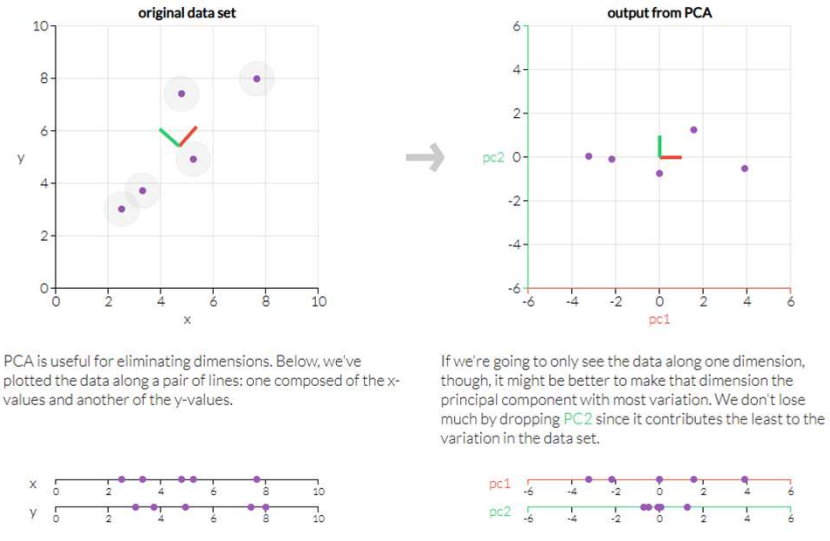
USTH 2025

$$\mathrm{PC}_x = X_{\text{centered}} \cdot \mathbf{x}$$

49

49

PCA Simulation



https://setosa.io/ev/principal-component-analysis/

50

50

Loss Function (Reconstruction Error)

Purpose

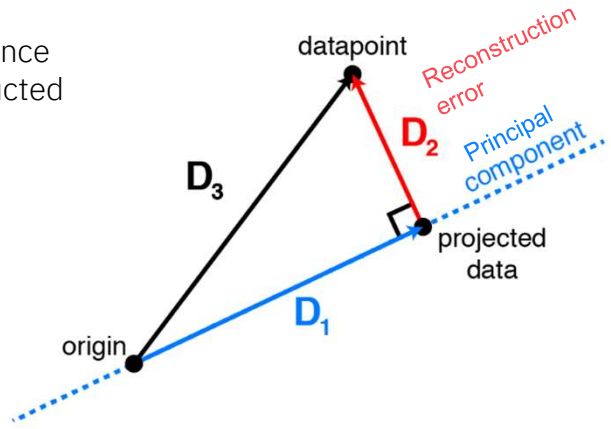
- **Quantifies information loss** caused by dimensionality reduction
- PCA selects components that minimize reconstruction error

Loss (Reconstruction error) measures the difference between the original data and the data reconstructed from the reduced components.

$$Loss = \frac{1}{n}\sum_{i=1}^{n}\left|x^i - \widehat{x^i}\right|^2$$

Where:

- xi = original data point
- $\widehat{x^i}$ = reconstructed data point
- n = number of samples



51

51

Cumulative variance in PCA

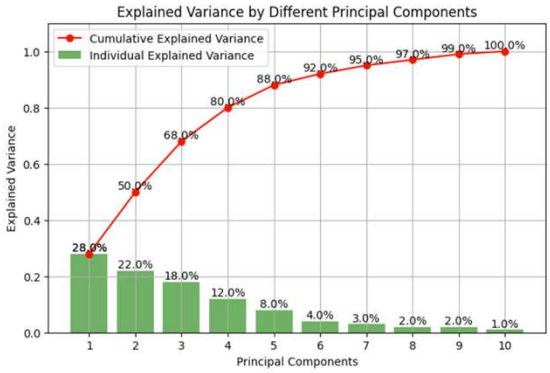
Cumulative explained variance measures how much of the total variability in the original data is captured by each principal component.

$$K = \frac{\sum_{i=1}^{k}\lambda_i}{\sum_{j=1}^{n}\lambda_j}$$

λ_i :→ How much information component j contains

$\sum_{i=1}^{k}\lambda_i$:→ Information kept by the first k components

$\sum_{j=1}^{n}\lambda_j$:→ Total information in the original data



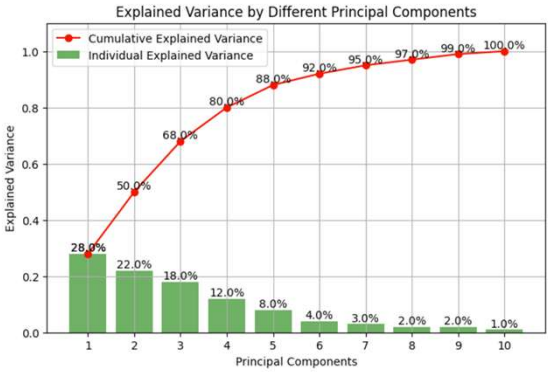
52

52

Choosing k explained

- First **3 components explain about 68%** of the variance, while **4 components reach ~80%,** showing rapid information gain early on.
- At **5 components capture ~88% of the variance**; beyond this point, each additional component adds very little new information (diminishing returns).
- A reasonable choice of **k is 5–6 components**, balancing information retention and model simplicity (elbow point).

Example Elbow plot



Balance simplicity and information

- Fewer components = simpler model
- More components = better information retention

USTH 2025

53

53

Case Study: Pizza Sale

Example: Pizza Dataset (7 features + brand/id)

The dataset refer to the amount of **water, protein, fat, ash, sodium, carbohydrates and calories** per 100 grams of the sample. (Example 6 first records)

	brand	id	mois	prot	fat	ash	sodium	carb	cal
1	A	14069	27.82	21.43	44.87	5.11	1.77	0.77	4.93
2	A	14053	28.49	21.26	43.89	5.34	1.79	1.02	4.84
3	A	14025	28.35	19.99	45.78	5.08	1.63	0.80	4.95
4	A	14016	30.55	20.15	43.13	4.79	1.61	1.38	4.74
5	A	14005	30.49	21.28	41.65	4.82	1.64	1.76	4.67
6	A	14075	31.14	20.23	42.31	4.92	1.65	1.40	4.67

Evaluate the correlation between all Features:

	mois	prot	fat	ash	sodium	carb	cal
mois	1	0.36	-0.171	0.266	-0.102	-0.592	-0.764
prot	0.36	1	0.498	0.824	0.429	-0.854	0.07
fat	-0.171	0.498	1	0.792	0.933	-0.64	0.765
ash	0.266	0.824	0.792	1	0.808	-0.899	0.326
sodium	-0.102	0.429	0.933	0.808	1	-0.62	0.672
carb	-0.592	-0.854	-0.64	-0.899	-0.62	1	-0.023
cal	-0.764	0.07	0.765	0.326	0.672	-0.023	1

USTH 2025

54

54

Case Study: Pizza Sale analysis with PCA

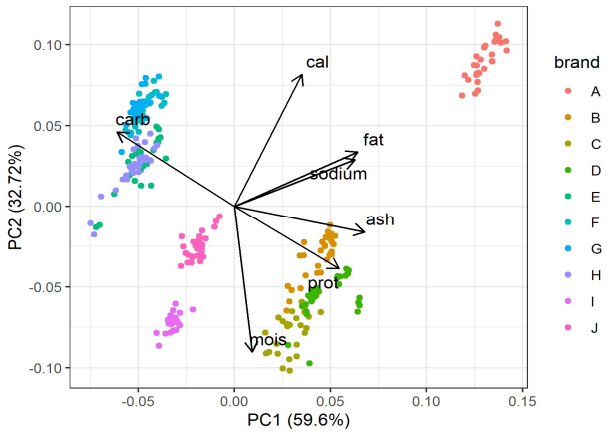
Principal Components:

- PC1 refers to the richness in fat, sodium, ash and protein, and the lack of carbohydrates.
- PC2 refers to the richness in calories and the lack of moisture.

	mois	prot	fat	ash	sodium	carb	cal
PC1	0.065	0.379	0.447	0.472	0.436	-0.425	0.244
PC2	-0.628	-0.27	0.234	-0.111	0.202	0.32	0.567

7 Dimensions data visualized on a 2D plane:

- If you want a **fatty and crispy pizza**, **Brand A** would be a good option.
- If you want a **soft and non-fat pizza**, **Brand I** would be a good choice.
- If you prefer a **balance of nutrients**, **Brand J** is the go-to.



USTH 2025

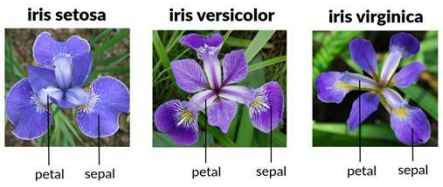
55

55

Lab 6.2: Dimensionality Reduction with Iris Dataset

Dataset

The Iris Dataset



Dimensionality Reduction

Problem

- Reduce the dimensionality (from 4 to 2 or 3) to visualize it

Features (4)

- Sepal length/ width
- Petal length/ width

Model

- Principal Component Analysis (PCA)

Lab Explained (about 1h)

- Step 1: Lab Setup
- Step 2: Perform Exploratory Data Analysis (EDA)
- Step 3: Feature Engineering
- Step 4: Model Training & Optimization
- Step 5: Outcomes

https://colab.research.google.com/drive/137B86RX5xv8UmGOxBVbQWfL9MyE5ChTR?usp=sharing

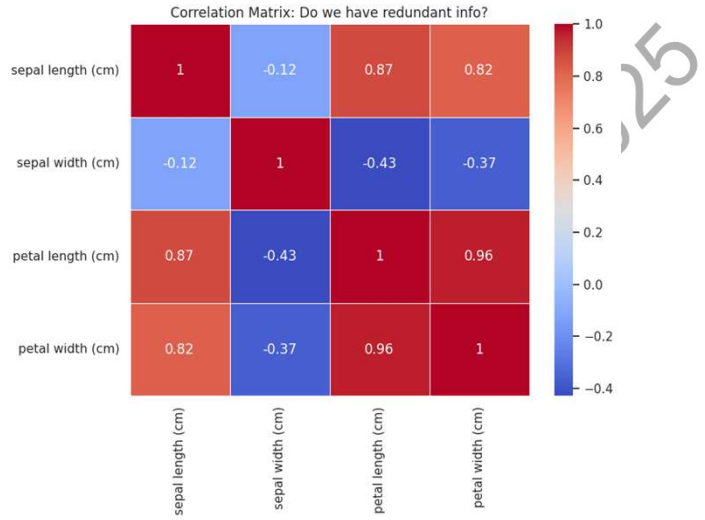
USTH 2025

56

56

Lab 6.2 – Perform Exploratory Data Analysis (EDA)

Feature Correlation: Calculate the correlation matrix



57

57

Lab 6.2 – Feature Engineering

Scale the data using Standardization scaler

```
# 1. Initialize Scaler
scaler = StandardScaler()

# 2. Fit and Transform
scaled_features = scaler.fit_transform(df)

# Check the new values (Mean should be ~0)
print("First 5 rows of Scaled Data:")
print(pd.DataFrame(scaled_features, columns=df.columns).head())
```

```
First 5 rows of Scaled Data:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0          -0.900681          1.019004          -1.340227         -1.315444
1          -1.143017         -0.131979          -1.340227         -1.315444
2          -1.385353          0.328414          -1.397064         -1.315444
3          -1.506521          0.098217          -1.283389         -1.315444
4          -1.021849          1.249201          -1.340227         -1.315444
```

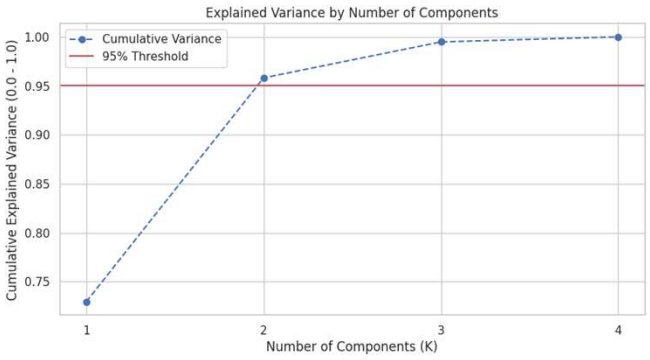
Since PCA measures importance using Variance, scaling ensures the algorithm finds true patterns rather than just picking the features with the biggest magnitude numbers.

58

58

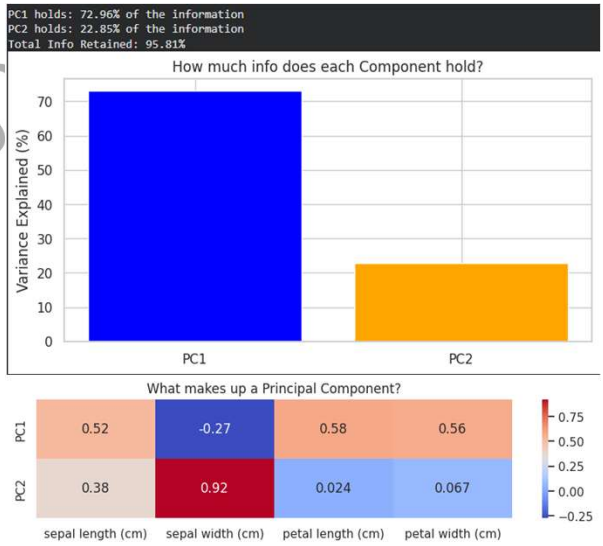
Lab 6.2 – Model Training & Optimization

Fit ALL the PCA components, get the Explained Variance to choose the best k-dimensions



Task: Modify then re-run the code with k-dimensions = 3 then visualize PC1, PC2, PC3

Choose k = 2, lets analyze:



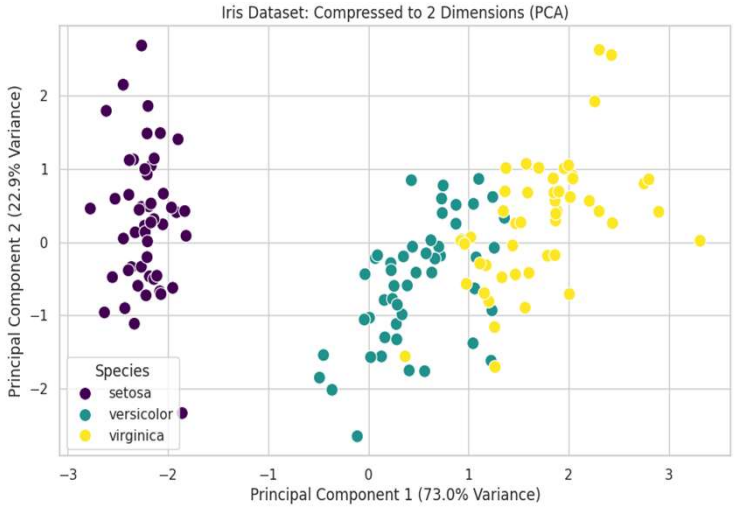
59

59

Lab 6.2 – Outcomes

Plot the 4 Dimension data into 2 Dimension

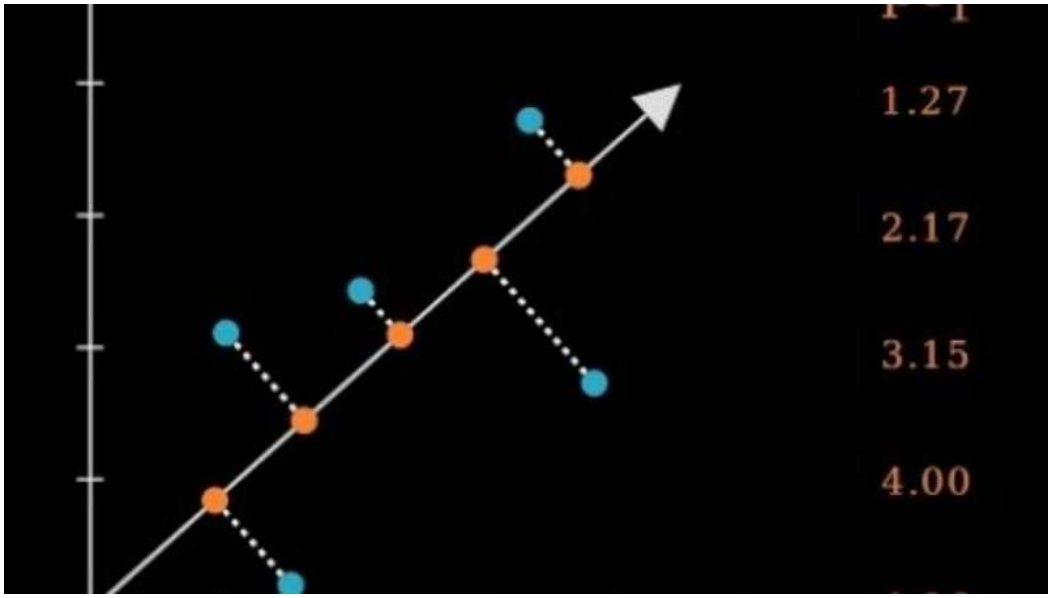
- We reduced the data by 50% (4D -> 2D) while maintaining clear separation, proving that half the original features can be removed.
- The model can learn to separate flower species without knowing the labels before.



60

60

PCA Video



https://www.youtube.com/watch?v=nEvKduLXFvk

61

61

Dimension Reduction: PCA— Key Takeaways

Pros

- Reduces high-dimensional data while preserving most variance
- Improves visualization and speeds up downstream models
- Removes noise and correlated features

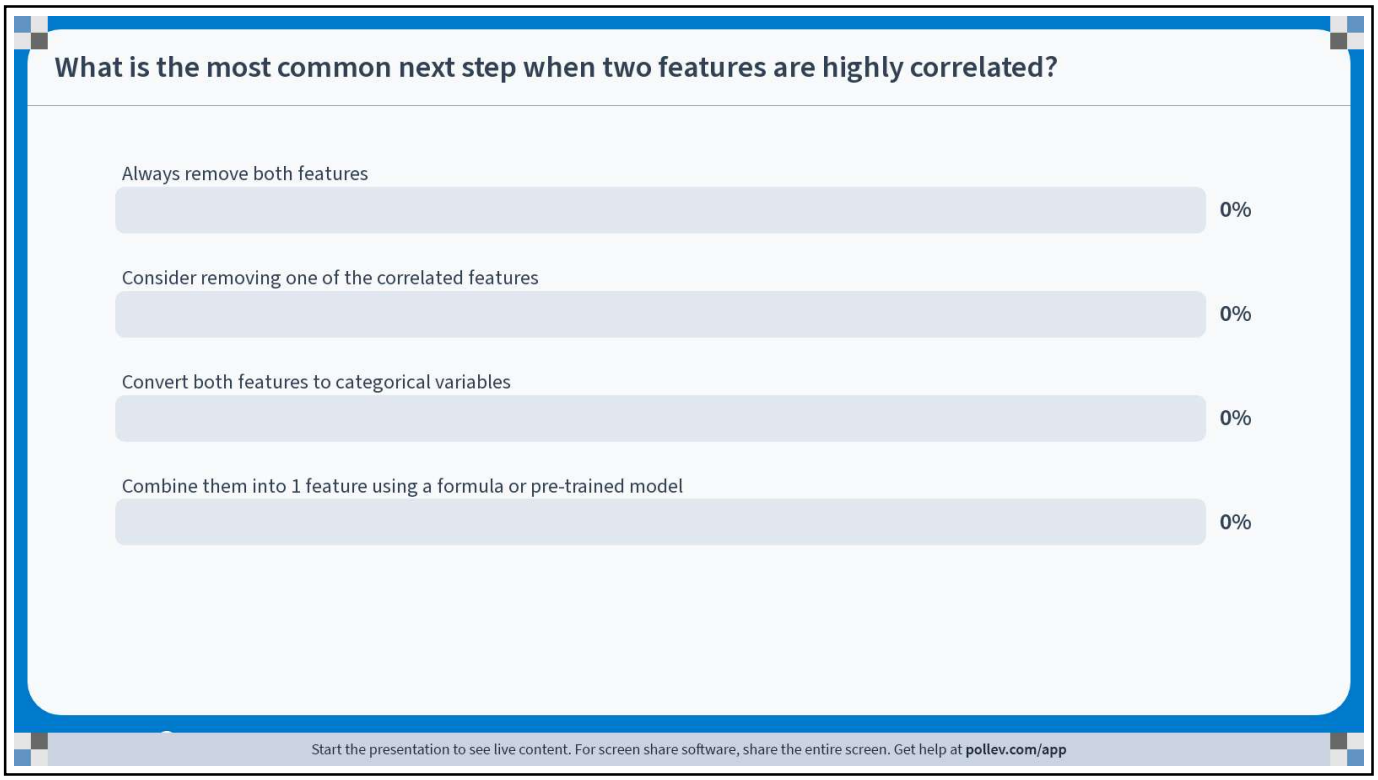
Cons

- Components are not easily interpretable
- Linear method (cannot capture non-linear structure)
- Information loss is unavoidable when reducing dimensions

USTH 2025

62

62



63

Thank you!

USTH 2025

64

64